

Lecture 7: Alignment w/ gap fn

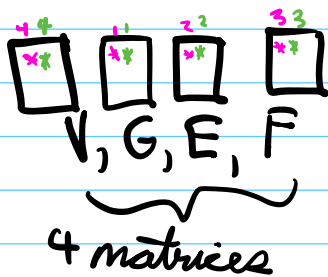
PowerPoint * See TA Sam Power Point

Inputs:

$X, Y, \alpha, \mu, \gamma, \tau$

- X, Y = strings of length m, n with characters indexed by i, j respectively
- α = match score, $\oplus$
- μ = mismatch penalty, $\ominus$
- γ = gap opening penalty $\ominus$
- τ = gap extension penalty (single-letter gap penalty)

Algorithm:



- 1) Fill the last 3
 - 2) Copy the result in the first matrix
- $V \Rightarrow$ general
 $G \Rightarrow$ diagonal
 $E \Rightarrow$ vertical
 $F \Rightarrow$ horizontal

$$\begin{aligned} V(i, 0) &= E(i, 0) = -\gamma - i\tau \\ V(0, j) &= F(0, j) = \gamma - j\tau \end{aligned}$$

$$V(i, j) = \max[E(i, j), F(i, j), G(i, j)]$$

$$G(i, j) = \begin{cases} V(i-1, j-1) + \alpha, & \text{if } x_i = y_j \\ V(i-1, j-1) - \mu, & \text{if } x_i \neq y_j \end{cases}$$

$$E(i, j) = \max[E(i, j-1), V(i, j-1) - \tau] - \tau$$

$$F(i, j) = \max[F(i-1, j), V(i-1, j) - \tau] - \tau$$

$3n$ (addition, multiplication, assignment)
 $O(m \cdot n)$ quadratic complexity

- ★ Can use alignment for Shazam to detect songs — sequences of notes
- Plagiarism, align 2 articles
- To transcribe words someone is saying
 - Hidden Markov models

Whiteboard

- If want global alignment with all $\oplus$ #'s, just add a positive constant w/ magnitude larger than those of all negative #'s
- Can't do for local alignment

Random Biosequences

Global optimal alignment of 2 random sequences of length N

Length of the optimal alignment $O(N)$

A	A	B
B	A	C
B	B	C
C	C	A

Random:

$$p(A) = \frac{4}{12}$$

$$p(B) = \frac{5}{12}$$

$$p(C) = \frac{3}{12}$$

LOCAL

$O(\log N)$



Q: How is it that it stops at length $O(\log N)$ for optimal local alignment?

A: The very special scoring scheme!

Karlin - Aetschul Statistics (1990)

↳ Done only for ungapped alignment $\{ = \{ =$

Ungapped pairs of max score ~~sites~~ of amino acids in

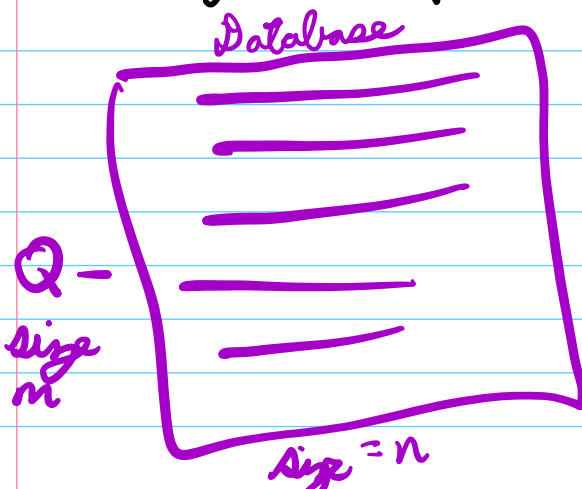
5 axioms for how to create the statistical model

- 1) Scoring matrix needs to have at least 1 positive score
- 2) The expected score of the scoring matrix needs to be negative
- 3) The letters of the random sequences are independent & identically distributed
- 4) Sequences are infinitely long
- 5) Alignments do not contain gaps

Karlin-Aetschul Equation:

$$E = kmme^{-\lambda S}$$

- E is the # of alignments expected by chance
- For alignments of score S



BLAST database search of size n & a query sequence of size m

Dayhoff LOD Score: log odds ratio

$$* LOD(\text{pair of amino acids}) = \log_2 \frac{\text{Obs. freq.}}{\text{Exp. freq. in the random model}}$$

$$R_{i,j} = \frac{q_{i,j}}{p_i p_j} \quad p_i, p_j = \text{frequencies of amino acids } i \& j$$

in amino acids
i & j

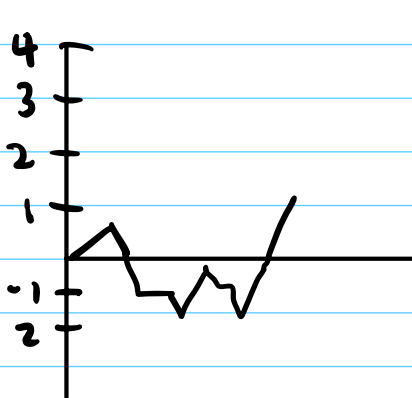
* If obs. freq. = exp. freq., then $LOD \approx 0$.

if $LOD > 0$, then pairing $i \& j$ in alignment is common

if $LOD < 0$, then pairing is unlikely

* $\begin{matrix} +1 & -1 \\ A & C & G & T & T & A \\ | & | & | & | & | & | \\ A & G & C & T & A & A \end{matrix}$

• Apply theory of random walks for alignment



Random walk